Memory-Efficient Training

Deep Dive into Kimi K3 — Chapter 39

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Foundation Books Project

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Goal: follow one versioned object through produce, transform, place, prefetch, consume, release — and prove a complete step fits, on a normalized fixture, without claiming any hardware measurement.

Roadmap

Memory is a live-set schedule

Policies that change lifetimes

Ownership and the complete step

Memory is a live-set schedule

The first OOM is an event, not a total

- Fixture *mem-k3-37-a*: capacity 32, fixed 18, naive bundle 13. Rank 0 plans $18 + 3(13) = 57$, rank 1 31; the first rejection is at event 3, at 44.

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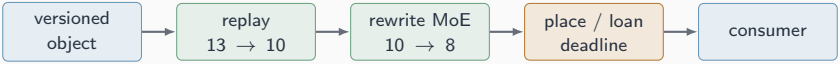
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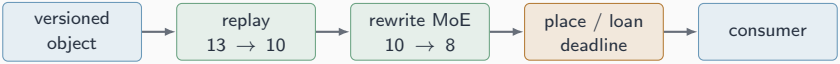
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- Objects carry full versioned keys; owners differ from holders; lossy storage names a metric and tolerance.

Policies that change lifetimes

Keep, replay, compress, offload, eliminate

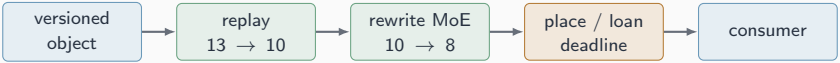


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- Compression ($8 \to 6$) is a separate arm: activation error $1/16$, local gradient error $3/32$, diagnostic $q = 113/4$. Offload trades capacity for readiness slack; remote loans move payload, not erase it.

Ownership and the complete step

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Device time, overlap, and training quality remain open. Next: Chapter 40's Multimodal Encoder and Input-Pipeline Optimization.